Question

A quantitative mixture of MnO_2, $NH_4H_2PO_4$, and H_3PO_4 is heated together to 300 $^\circ C$ (Reaction 1). After washing with water and drying, a purple powder $\mathbf{A}$ is obtained, which contains no water of crystallization and has a molecular weight of 246.9. If it is dissolved in dilute sulfuric acid, the solution color does not change significantly, but a precipitate is formed (Reaction 2), while dissolving it in concentrated sulfuric acid yields a clear red solution.

Researchers performed thermogravimetric analysis on $\mathbf{A}$, and the intermediate products exhibited a variety of colors (the weight loss ratios here are all theoretical values): $\mathbf{A}$ is heated to 340 $^\circ C$ in an inert atmosphere, with a weight loss of 3.65%, yielding a blue substance (Reaction 3); continuing to heat to 460 $^\circ C$, it loses another 10.14% of its weight, yielding a white substance $\mathbf{B}$ (Reaction 4); subsequently, heating to 800 $^\circ C$ does not result in any noticeable weight loss, $\mathbf{B}$ melts and completely transforms into a pink liquid $\mathbf{C}$. The formula weight of $\mathbf{C}$ is 1.5 times that of $\mathbf{B}$.

The following statements exist:

1. The sum of the coefficients on the right side of the chemical equation for Reaction 1 is 15.

2. The sum of the coefficients on the left side of the chemical equation for Reaction 2 is less than 8.

3. The sum of all the atoms in the chemical formula of the blue product obtained in Reaction 3 is 27.

4. The sum of the coefficients on both sides of the chemical equation for Reaction 4 is less than or equal to 12.

5. The ideal point group of the anion of $\mathbf{C}$ is C_{3v}.

The option containing all the correct statements is:

A. All other options are incorrect

B. 2

C. 5

D. 4

E. 1,2,3,4

F. 2,3,5

G. 1,2,4

H. 2,4,5

I. 1,3,5

J. 2,5

K. 1,3

L. 3,4,5

M. 1,2,4,5

N. 2,3

O. 1,5

P. 4,5

Q. 3,5

R. 1

S. 1,2,3,5

Answer

Correct Answer: I

Detailed Explanation

$\mathbf{A}$ should contain at least 1 Mn, 1 NH_4^+ and 1 PO_4^{3-}. Subtracting these from the molecular weight of $\mathbf{A}$, 246.9, leaves approximately 79, which is $P + 3O$. Therefore, $\mathbf{A}$ is $NH_4MnP_2O_7$.

CHECKPOINT 1 PTS

$\mathbf{A}$ is $NH_4MnP_2O_7$

Reaction 1: Mn is reduced, and the only element that can be oxidized is O, producing O_2. The equation is: $4MnO_2 + 4NH_4H_2PO_4 + 4H_3PO_4 \rightarrow 4NH_4MnP_2O_7 + 10H_2O + O_2$. The sum of the coefficients on the right side is 15, so statement 1 is correct.

CHECKPOINT 1 PTS

Reaction 1 equation is: $4MnO_2 + 4NH_4H_2PO_4 + 4H_3PO_4 \rightarrow 4NH_4MnP_2O_7 + 10H_2O + O_2$

Reaction 2: Dissolving in dilute sulfuric acid produces a precipitate, corresponding to the disproportionation of trivalent Mn to produce Mn^{2+} and MnO_2. The equation is: $2NH_4MnP_2O_7 + 2H_2SO_4 + 4H_2O \rightarrow (NH_4)_2SO_4 + MnSO_4 + MnO_2 + 4H_3PO_4$. Statement 2 is incorrect.

CHECKPOINT 1 PTS

Reaction 2 equation is: $2NH_4MnP_2O_7 + 2H_2SO_4 + 4H_2O \rightarrow (NH_4)_2SO_4 + MnSO_4 + MnO_2 + 4H_3PO_4$

First step weight loss: $M_1 = w_1 \times M = 3.65 \times 10^{-2} \times 246.922\ g \cdot mol^{-1} = 9.01\ g \cdot mol^{-1}$, corresponding to 0.5 H_2O molecules.

CHECKPOINT 2 PTS

First step weight loss corresponds to 0.5 H_2O molecules

Second step weight loss: $M_2 = w_2 \times M = 10.14 \times 10^{-2} \times 246.922\ g \cdot mol^{-1} = 25.04\ g \cdot mol^{-1}$, corresponding to $N + 3H + 0.5O$ (what is lost may be N_2, H_2O, NH_3).

CHECKPOINT 2 PTS

Second step weight loss corresponds to $N + 3H + 0.5O$, what is lost may be N_2, H_2O, NH_3

Therefore, in the two steps of weight loss, 1 molecule of $NH_4MnP_2O_7$ loses 1 N, 4 H, and 1 O, corresponding to the simplest formulas of $\mathbf{B}$ and $\mathbf{C}$ being MnP_2O_6.

CHECKPOINT 2 PTS

$\mathbf{B}$ and $\mathbf{C}$ simplest formulas are MnP_2O_6

Since the formula weight of $\mathbf{C}$ is 1.5 times that of $\mathbf{B}$, we can deduce from the simplest case that $\mathbf{B}$:

$Mn_2P_4O_{12}$, $\mathbf{C}$: $Mn_3P_6O_{18}$

CHECKPOINT 1 PTS

$\mathbf{B}$: $Mn_2P_4O_{12}$, $\mathbf{C}$: $Mn_3P_6O_{18}$

Reaction 3: Two molecules of $\mathbf{A}$ produce one molecule of H_2O, and also produce $Mn_2(NH_3)_2P_4O_{13}$. The equation is: $2NH_4MnP_2O_7 \rightarrow Mn_2(NH_3)_2P_4O_{13} + H_2O$. Statement 3 is correct.

CHECKPOINT 1 PTS

Reaction 3 equation is: $2NH_4MnP_2O_7 \rightarrow Mn_2(NH_3)_2P_4O_{13} + H_2O$

Reaction 4: $Mn_2(NH_3)_2P_4O_{13}$ is converted to $Mn_2P_4O_{12}$, losing NH_3, N_2 and H_2O. The equation is: $3Mn_2(NH_3)_2P_4O_{13} \rightarrow 3Mn_2P_4O_{12} + 4NH_3 + N_2 + 3H_2O$. Statement 4 is incorrect.

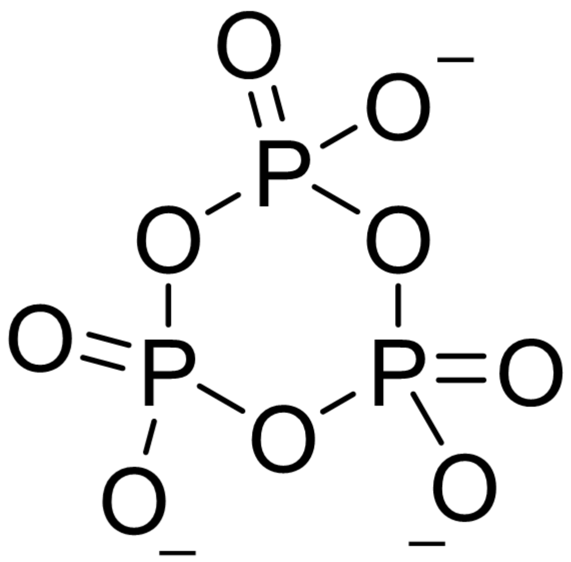
CHECKPOINT 1 PTS

Reaction 4 equation is: $3Mn_2(NH_3)_2P_4O_{13} \rightarrow 3Mn_2P_4O_{12} + 4NH_3 + N_2 + 3H_2O$

The anion of $\mathbf{C}$ is $P_3O_9^{3-}$, and its structure is O=P(OP(O1)([O-])=O)([O-])OP1([O-])=O, the ideal point group is C_{3v}. Statement 5 is correct.

CHECKPOINT 1 PTS

C anion is $P_3O_9^{3-}$, ideal point group is C_{3v}



C anion structure is O=P(OP(O1)([O-])=O)([O-])OP1([O-])=O

Statements 1, 3, and 5 are correct, choose I